Neven Caplar

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Professional Summary

Scientific data systems lead with 9+ years of experience building data platforms for large astronomical surveys. Lead the open-source HATS/LSDB format and toolkit for cross-matching and analyzing billion-row catalogs, used for data from Rubin Observatory and other large surveys. Co-PI on a U.S. Department of Energy (DOE) Genesis Mission project building a multimodal data service for AI model training.

Experience

University of Washington / LINCC Frameworks *2022 – Present*
Research Scientist/Engineer

- Scientific lead and project owner of HATS/LSDB: a survey-agnostic, Parquet-based format and Python toolkit for scalable analysis and cross-matching of billion-object catalogs.
- Own the technical roadmap and guide architecture for distributed (Dask) catalog analysis; extending the platform to upcoming surveys (Roman, Euclid, Gaia DR4, Schmidt Observatory System).
- Co-PI, DOE Genesis Mission award (with Carnegie Mellon University and SLAC): unified multi-modal data service (catalogs, images, spectra) and streaming data loaders for AI model training.
- Lead a team of 8 software engineers and scientists (UW, CMU); coordinate development with NASA archives (STScI/MAST, IPAC/IRSA), international data centers, and survey teams.

Vera C. Rubin Observatory *2022 – 2025*
Project Scientist

- Contributed to commissioning and validation of time-domain pipelines, including difference imaging and large-scale light-curve construction.
- Defined scientific requirements and validation strategies for pipelines processing nightly multi-terabyte data volumes; co-author of the Data Preview 1 paper.

Princeton University *2017 – 2022*
Associate Professional Specialist

- Built the 2D point-spread-function model for the Prime Focus Spectrograph (Subaru Telescope) data reduction pipeline, combining physical modeling and machine-learning methods.

Education

Ph.D., Science, Swiss Federal Institute of Technology (ETH) Zurich 2017
M.Sc., Physics, University of Zagreb 2010

Research

- 22 peer-reviewed publications (7 first-author) plus 2 submitted; ∼1,960 citations
- First author of the LSDB paper (ADASS 2024): large-scale catalog cross-matching and analytics
- Ph.D. thesis on AGN population evolution; variability study of 28,000 AGN (2.4M measurements)

Skills

Leadership: project leadership, roadmap ownership, cross-institutional coordination, grant writing
Data & Software: Python, Dask, Parquet/PyArrow, pandas, JAX, HPC (SLURM), cloud storage
Domains: large catalogs, time-domain surveys, data validation, data for AI/ML